

Install from source.

01

One verified script prepares, builds, checks, and starts the local desktop app.

BEFORE YOU BEGIN

Use Terminal on Mac or PowerShell on Windows as a normal user. The first setup needs internet access for application dependencies and a private Node.js runtime when a compatible system copy is unavailable.

Mac - Terminal

```
git clone https://github.com/adammalin/Map-Maker-Studio.git \  
"$HOME/Map-Maker-Studio"  
cd "$HOME/Map-Maker-Studio"  
/bin/zsh scripts/setup-macos.zsh
```

Windows - PowerShell

```
git clone https://github.com/adammalin/Map-Maker-Studio.git `  
"$env:USERPROFILE\Map-Maker-Studio"  
Set-Location "$env:USERPROFILE\Map-Maker-Studio"  
powershell.exe -NoLogo -NoProfile -ExecutionPolicy Bypass `  
-File ".\scripts\setup-windows.ps1"
```

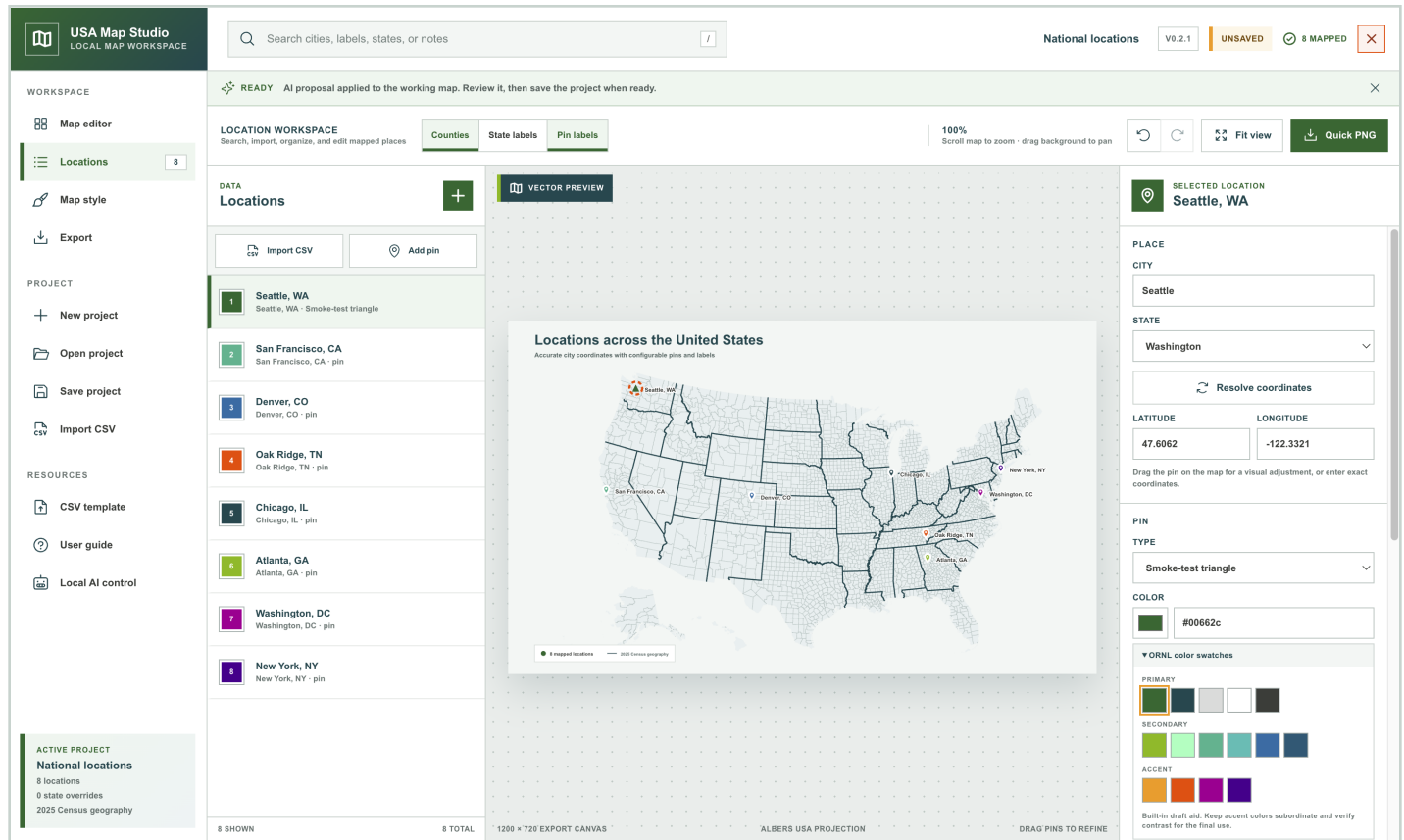
WHAT SETUP DOES

The script installs exact dependencies, builds the Electron interface, and runs a hidden map/UI smoke check. It does not create a DMG, PKG, MSI, or EXE installer, disable Gatekeeper, or make system-wide changes. Later, double-click Start-USA-Map-Studio.command on Mac or Start-USA-Map-Studio.cmd on Windows.

Build the map visually.

A focused canvas, location list, and inspector keep every change visible.

02



1 MAP EDITOR

Work on the largest canvas view. Select pins directly on the map and refine coordinates, labels, symbols, and colors in the inspector.

2 LOCATIONS

Open the searchable location manager to import CSV data, add pins, organize the full list, and edit the selected place.

3 MAP STYLE

Focus on global geography and state appearance: fills, boundaries, county lines, label halo, abbreviations, and legend.

MAP CONTROLS

Use the mouse wheel to zoom, drag the map background to pan, and choose Fit view to reset. Drag a pin for a visual adjustment; enter or resolve coordinates for an exact location. Editor selection outlines never appear in an export.

Bring data in. Keep it whole.

03

Flexible columns feed a versioned project file that can be saved and reopened.

MINIMUM CSV

Two columns are enough

```
city,state  
Oak Ridge,TN  
Seattle,WA
```

Or supply exact coordinates

```
city,state,latitude,longitude  
Denver,CO,39.7392,-104.9903
```

OPTIONAL VISUAL FIELDS

- **Labels**
label, show_label, label_color, and label_position (right, left, above, or below).
- **Pins**
pin_type (pin, circle, square, diamond, or star), pin_color, and pin_size from 6 through 40. Custom SVG pins are added in the inspector.
- **Additional data**
notes plus any custom columns. Unknown columns stay in customData and return with the project.
- **Review before adding**
The import summary separates ready rows from unresolved or malformed rows. Choose Add or Replace deliberately.
- **Save project**
Save project creates a .usmap.json file containing map styling, every location, embedded custom SVG pins, labels, custom data, schema, and timestamps.
- **Open project**
Opening validates the schema and required fields before replacing the canvas. Unsupported or unrelated JSON is rejected.

COORDINATE RULE

When latitude and longitude are blank, the app matches city and state to one of 32,350 representative points from the 2025 U.S. Census National Places Gazetteer. If both coordinates are supplied, the app preserves them. An unresolved place stays out of the map and appears in the import issue list so it cannot be plotted silently at the wrong location.

Export what you see.

04

SVG, PNG, PowerPoint, and project JSON share one composition.

1 SVG

Best for scalable artwork and downstream design tools. State lines, labels, pins, and legend remain vector content.

2 PNG

A 2400 x 1440 raster image for documents, email, and quick review.

3 PPTX

A one-slide 16:9 PowerPoint with the map embedded as scalable vector artwork and source notes.

4 JSON

The complete editable project, including embedded custom pin artwork. Reopen it to continue work.

GOOD WORKING SEQUENCE

- 1 Save the project**
Keep a .usmap.json source file before making presentation-specific variations.
- 2 Export the audience copy**
Choose SVG, PNG, or PowerPoint after checking title, labels, state colors, and legend at Fit view.
- 3 Keep source and output together**
Store the CSV and project JSON beside exported artwork when the map must be reproduced later.

TROUBLESHOOTING

Setup fails: rerun the setup script; it repairs dependencies and rebuilds safely. A city is unresolved: add exact latitude and longitude or use the official Census place name. Labels overlap: select a location and move its label left, right, above, or below. The app is not built: run the setup script instead of the start shortcut.

LOCAL DATA BOUNDARY

Normal editing, place lookup, map rendering, and export run locally. USA Map Studio does not send CSV or project content to a service. Files are written only to locations you choose. The bundled map and place data come from U.S. Census Bureau 2025 Cartographic Boundary and Gazetteer files.

Let AI prepare. You decide.

05

ChatGPT desktop, Codex, and compatible local clients can stage map changes for review.

1 KEEP THE APP OPEN

The setup script registers `usa_map_studio`. Restart the AI client, open USA Map Studio, then use `/mcp` to confirm the connection.

2 ASK FOR A DRAFT

The AI reads the named open project, uses its current timestamp, and stages one proposal. No map or saved file changes yet.

3 REVIEW IN THE APP

Compare Before and After. Apply to the working map or Reject. Save the project separately only after checking the canvas.

MANUAL CONNECTION COMMANDS

Register or repair

```
npm run mcp:install
# Restart ChatGPT desktop or Codex, then use /mcp
```

Remove managed connection

```
npm run mcp:remove
# Existing unrelated MCP settings remain in place
```

WHAT THE AI CAN DO

Read

Check app status, read the complete current project, list locations, and validate project JSON.

Prepare

Stage CSV imports, exact locations, custom SVG pins, location edits, removals, map styling, or a complete project replacement.

Cannot bypass review

Write tools create one pending proposal. They cannot click Apply, write an export, or silently save project JSON.

Stale changes stop

If the working map changes after the AI reads it, the proposal must be rebuilt from the current project.

PRIVACY BOUNDARY

The bridge listens only on this computer and uses a new random token for each app launch. Project data stays local until a read tool returns it to the connected AI conversation. Use only an AI client approved for the map content. ChatGPT web cannot directly reach this private desktop bridge; that would require a separate hosted plugin.

Import a pin. Reuse the palette.

06

SVG artwork stays with the project, and named ORNL swatches stay one click away.

CUSTOM SVG PIN WORKFLOW

1 IMPORT

Select a location, choose Import custom SVG pin, and select a vector-only .svg file up to 500 KB.

2 COLOR

Use `currentColor` in the SVG when the design should follow each location's pin color. Otherwise, embedded SVG fills are preserved.

3 SAVE

The sanitized SVG is stored inside `.usmap.json`. Move or share that one project file without losing the pin artwork.

PRIMARY



ORNL Green
#00662C



Hale Navy
#00454D



Graphite
#DBDCDB



Polar
#FFFFFF



Dark Matter
#373A36

SECONDARY



Energy
#7DBA00



Mist
#8BFEBF



Biome
#00B38F



Aqua
#00BDB5



Infinity
#006BA6



Hydro
#005776

ACCENT



Forge
#FF9E1B



Spark
#FE5000



Plasma
#B50094



Pulsar
#4E008E

SAFE BY DESIGN

The importer removes scripts, event handlers, external references, embedded raster images, styles, and unsupported elements before display or save. ORNL swatches are a draft-production aid: check contrast, use accent colors sparingly, and complete the normal communications review before release.